3SUM Is Really Hard: A Real-to-Integer Reduction

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Bocconi

FIS

Ministero dell'Università e della Ricerca

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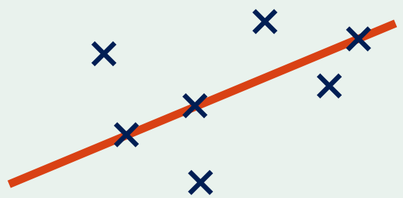
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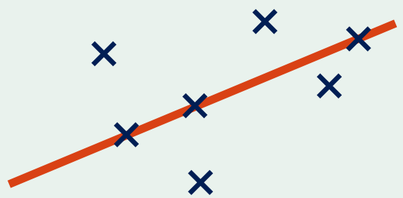
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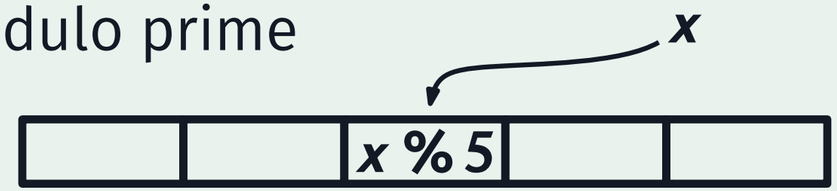


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3SUM originates from geometric problems.



- To better understand the **algorithmic techniques**!
- **Integer 3SUM Techniques:**
  - Hashing modulo prime



  - Bit tricks

$x = 101\,0110$

- Are they required?
- Common pattern: Restrict algorithm, **force new techniques**

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Theorem [Chan, Vassilevska Williams, Xu '22]

If Integer All-Nums 3SUM in time $O(n^{1.2-\epsilon})$, then Real 3SUM in time $O(n^{2-\delta})$.

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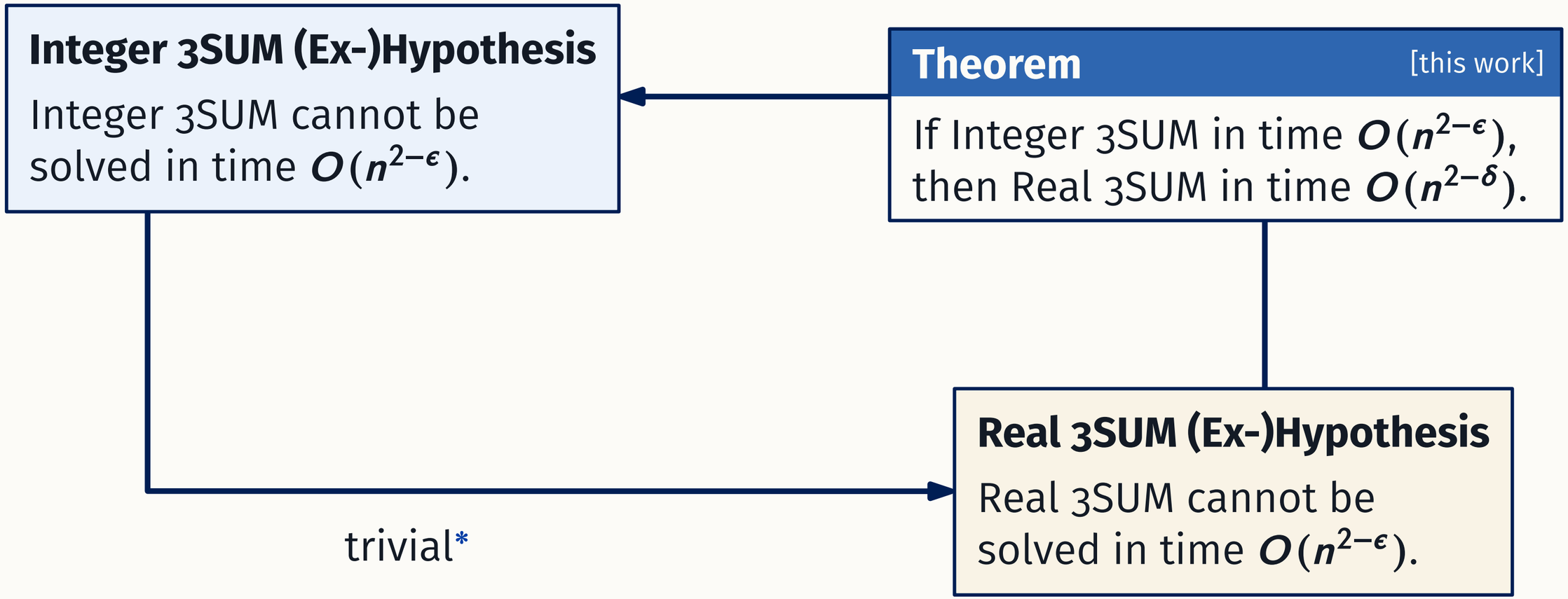
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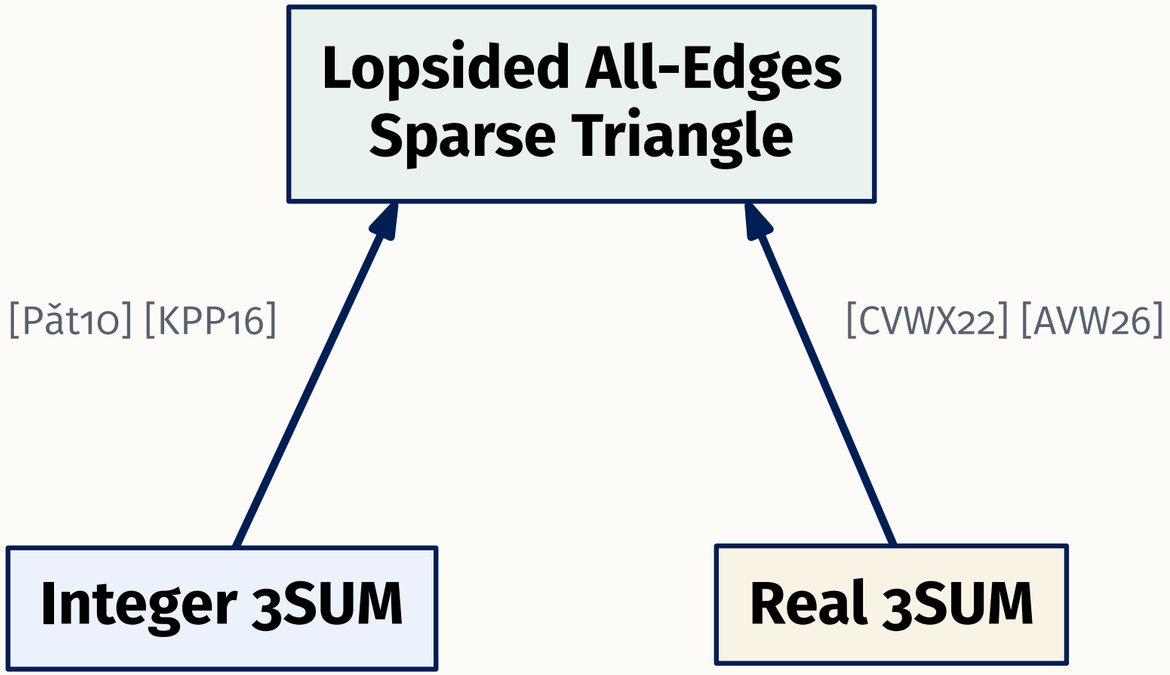
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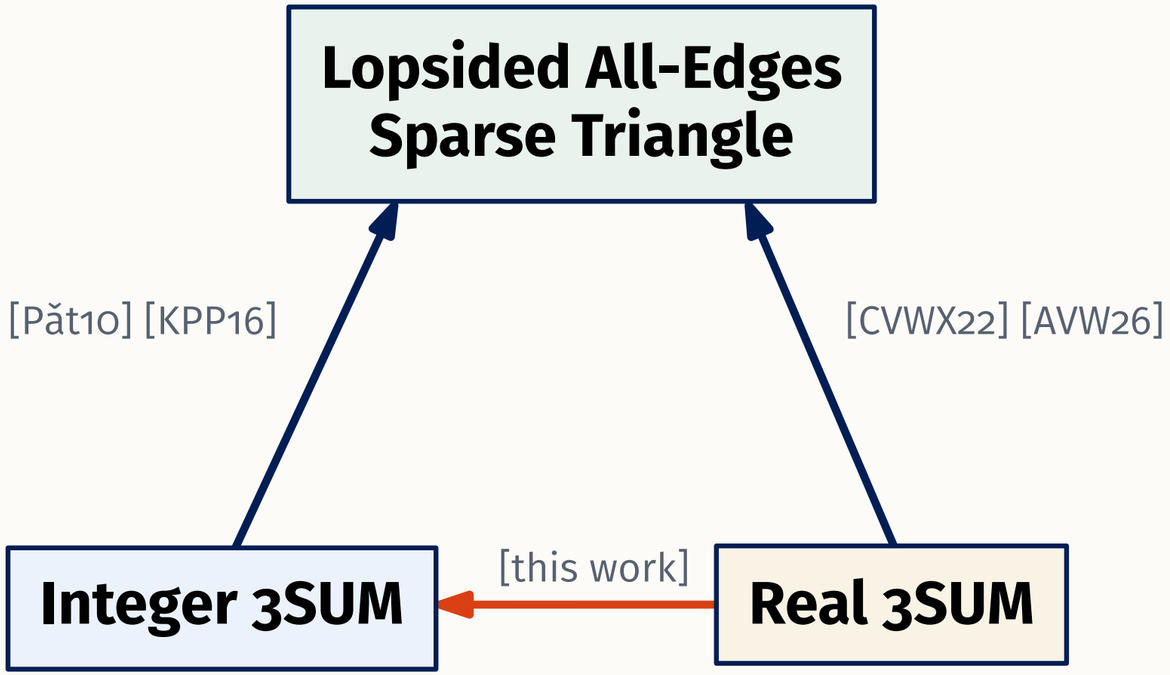
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Issue:

- Integers are too big
- Running time of FT: $O(n^6)$ (uses LLL internally)

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α = **subset sum cover** of A

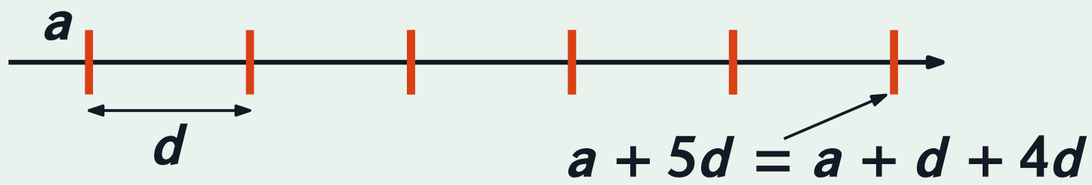
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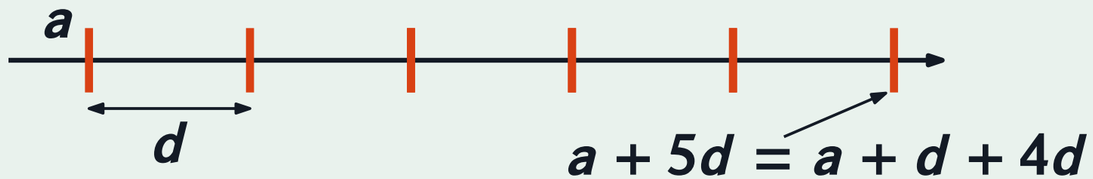


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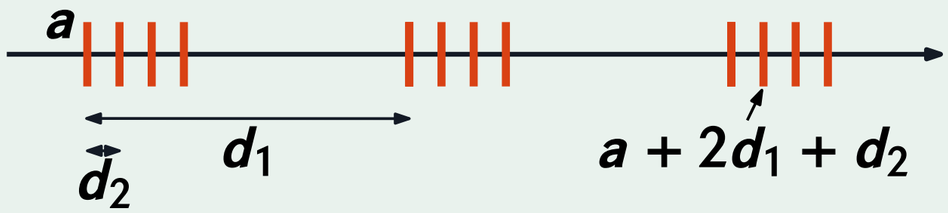
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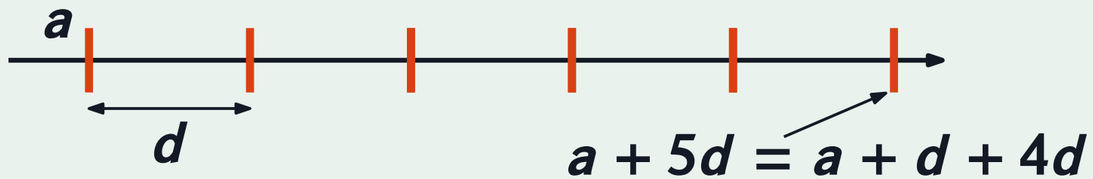
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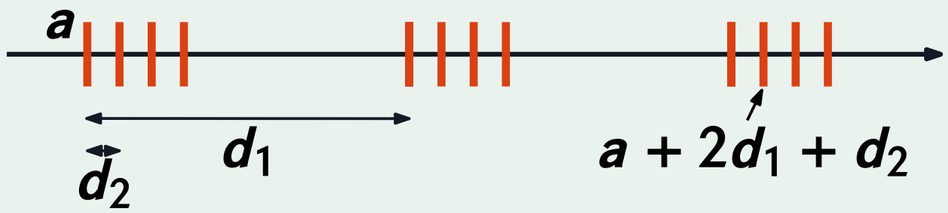
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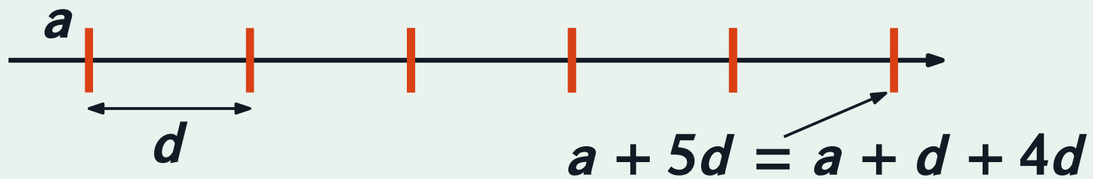


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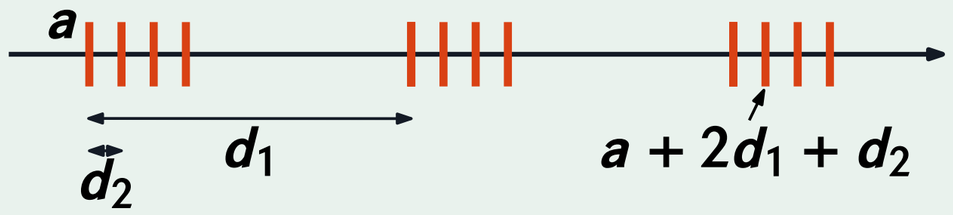
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Freiman's Theorem [Freiman '66] [Ruzsa '94]

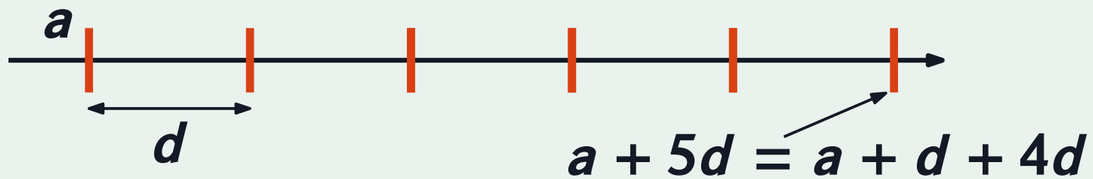
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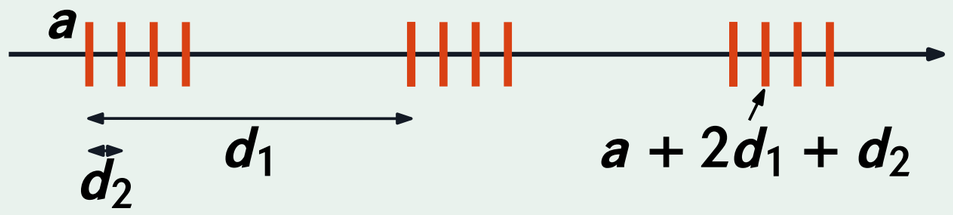
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- **For reals**, we give completely new proof:

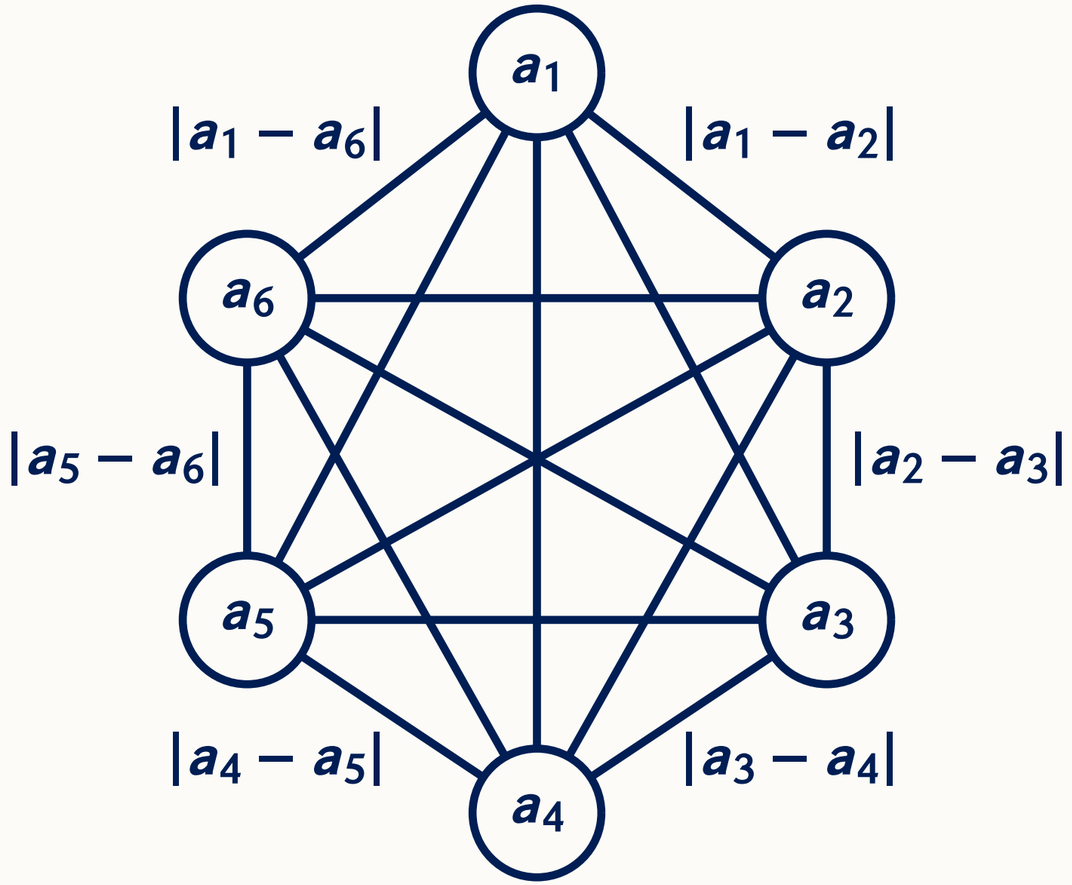
"Poor Man's" Freiman's Theorem [this work]

Let $A \subseteq \mathbb{R}$ have doubling $|A + A| \leq K|A|$.

Then, we **can quickly construct** a subset sum cover of A of size $(K \log |A|)^{O(1)}$.

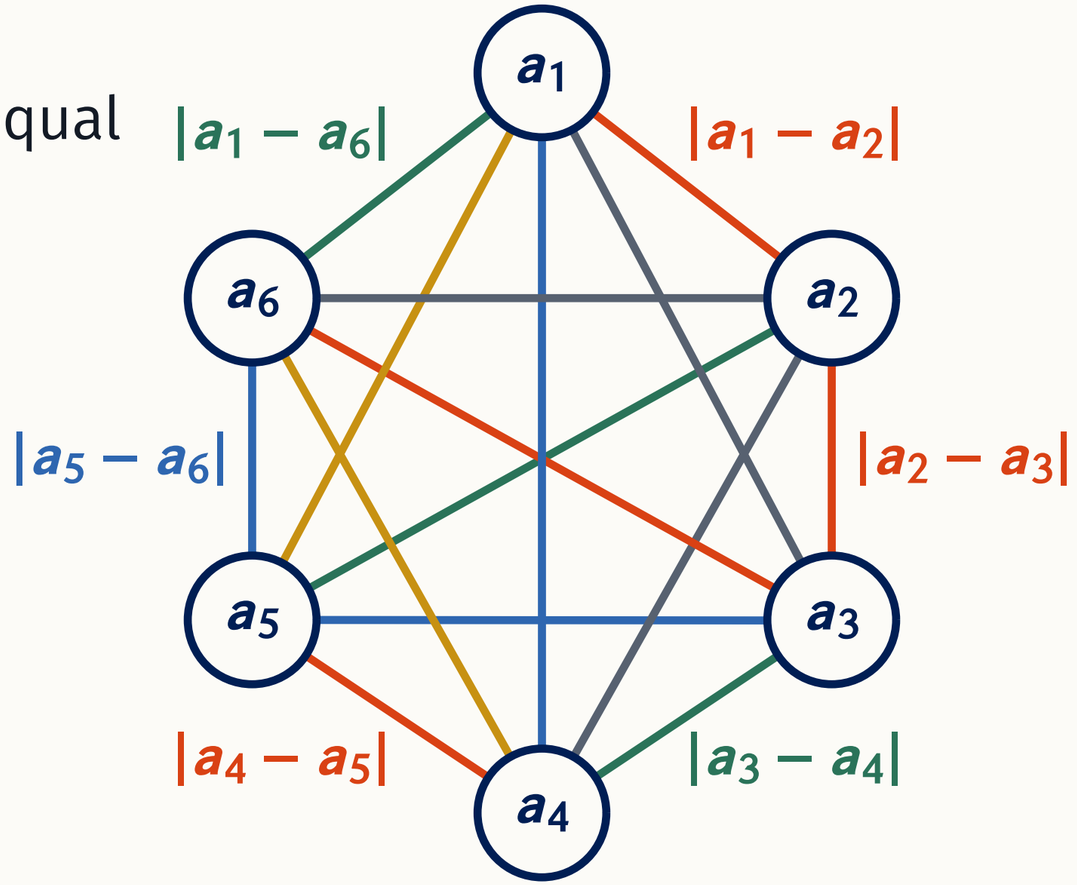
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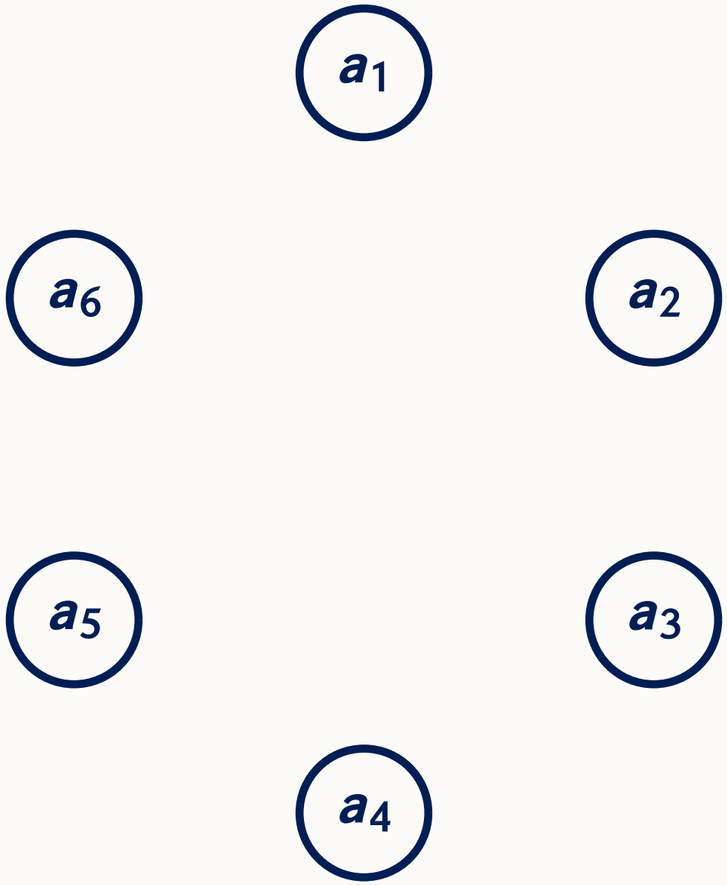
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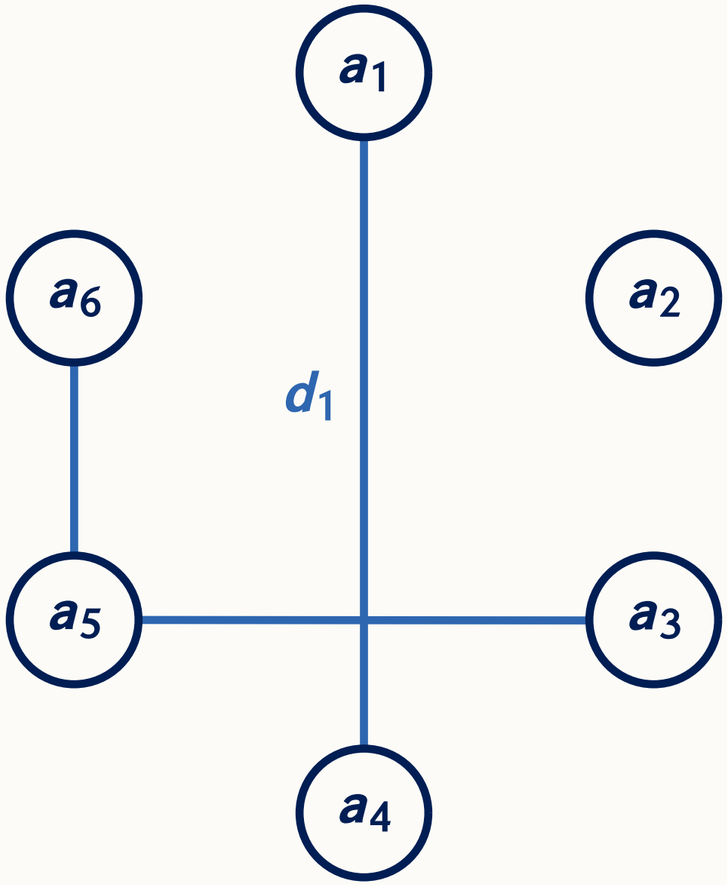
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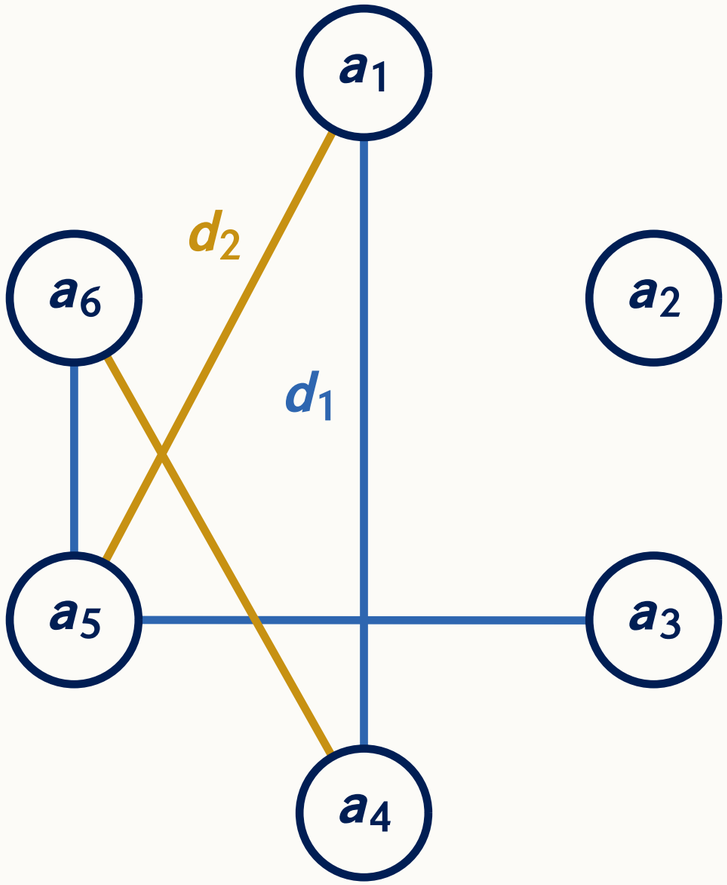
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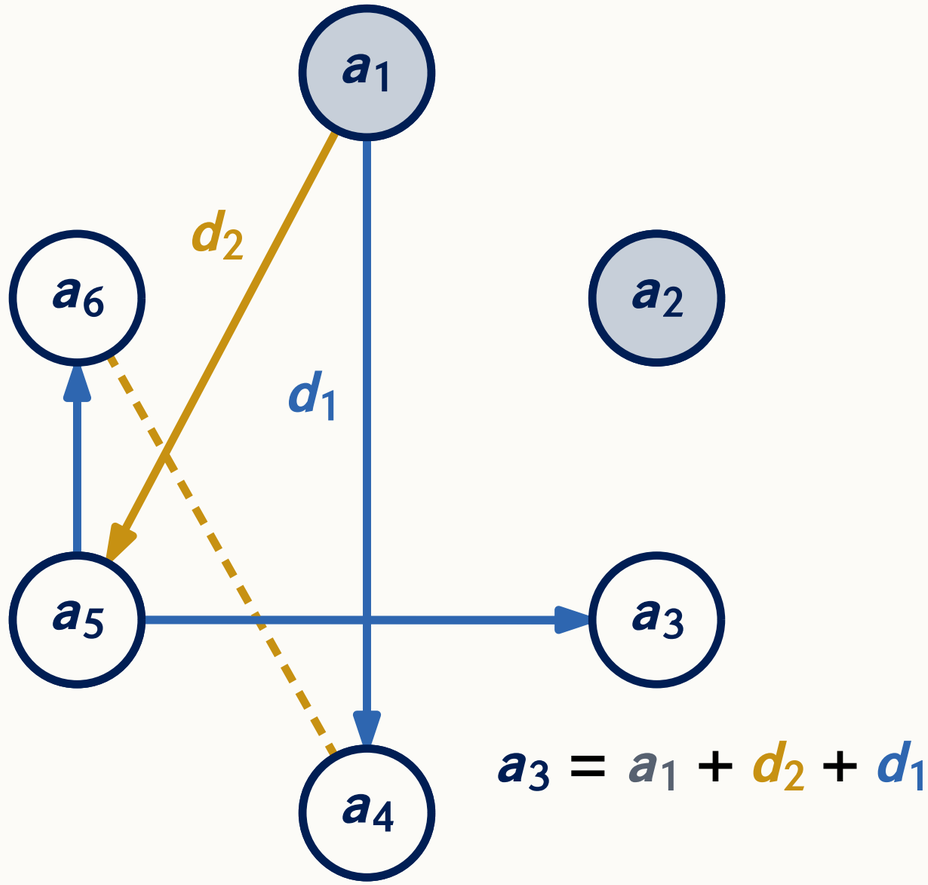
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Subset Sum Cover: $\alpha = (a_1, a_2, d_1, d_2)$

(Add $2d_i, 4d_i, \ldots, 2^{\lfloor \log |A| \rfloor} d_i$ and negative differences in general)

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Algorithm: Real 3SUM

Input: $A \subseteq \mathbb{R}$



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Apply **Frank–Tardos** to A

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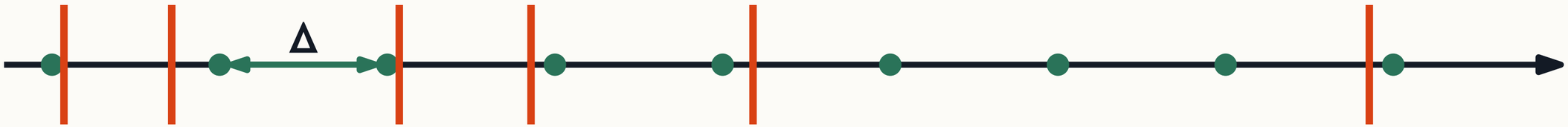
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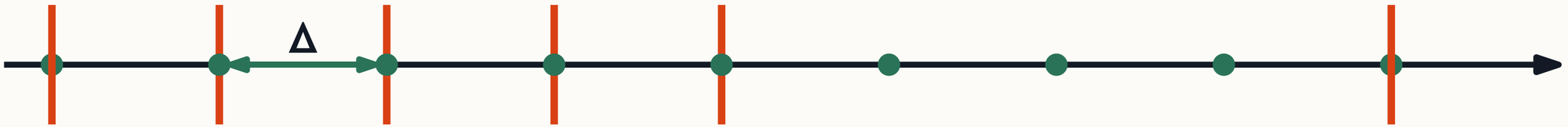
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Idea: Round to create structure!



*not a valid operation.
We round only *symbolically*.

Real-to-Integer 3SUM

Algorithm: Real 3SUM **Input:** $A \subseteq \mathbb{R}$

Round* numbers s.t. there are $\Theta(n^{2-\delta})$ solutions

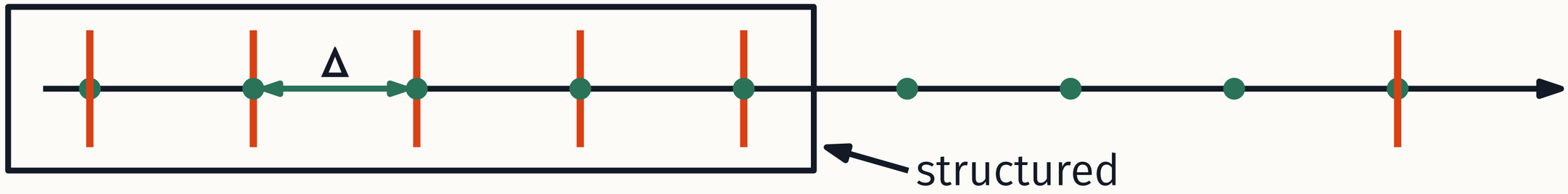
$\Longrightarrow \exists$ structured part $A' \subseteq A$

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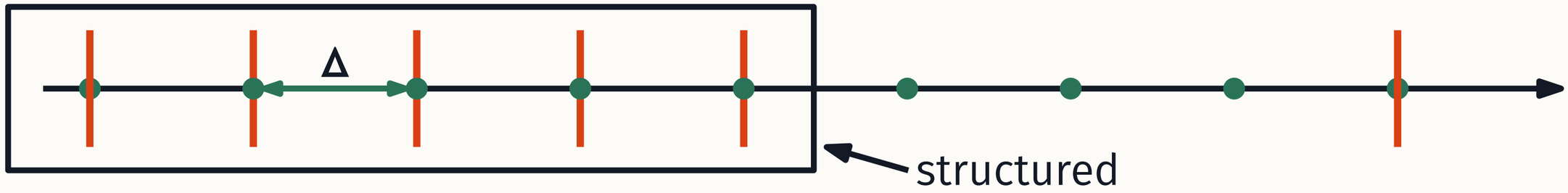
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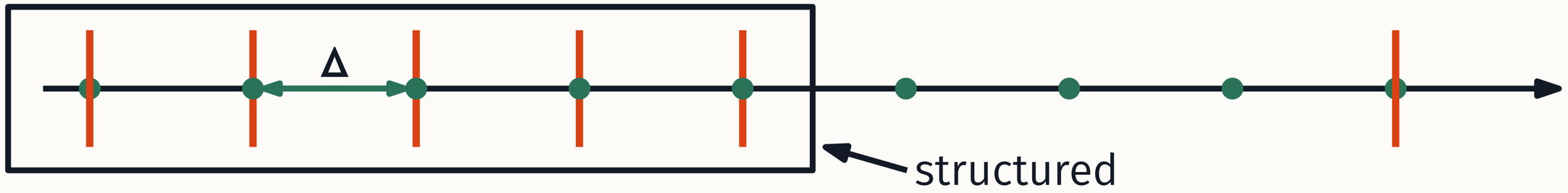
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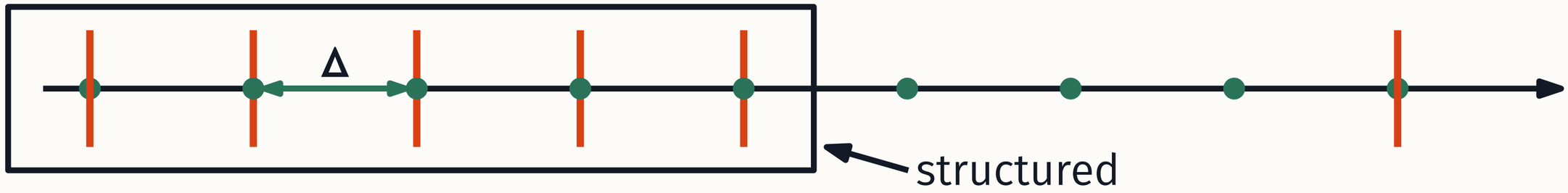
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Theorem [this work]

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Real-to-Integer 3SUM

Thank you!

Algorithm: Real 3SUM **Input:** $A \subseteq \mathbb{R}$

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